

DAP Summing Stage Rev3

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Attribution

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Part Identification

One differential summing stage per channel converts the two I/V outputs (e.g. DACR and DACRB legs) to a single-ended signal and applies a second-order low-pass. Topology: differential multiple-feedback (MFB) low-pass, Butterworth response.

Signal chain per channel: I/V ($\times 2$) $\rightarrow$ **differential summing + MFB low-pass** $\rightarrow$ OPA1622 $\rightarrow$ headphone jack.

Op-amp: OPA1612AIDRGR (TI, dual, 8-SON). One package covers both channels.

Design Requirements

- **Gain:** 0.44, from 0.85 V peak (0.6 V RMS) headphone target $\div$ 1.93 V peak nominal differential input. The OPA1622 stage is unity gain.
- **Response:** Butterworth, $Q = 0.7071$. Passband ripple 0 dB.
- **Corner frequency:** $f_0 = 70$ kHz. Minimum for ≤ 0.05 dB loss at 20 kHz is 60.96 kHz; 70 kHz leaves margin for component tolerance.
- **Chain budget:** total loss at 20 kHz ≤ 0.1 dB (I/V stage + summing stage).

Design Constraints

- ± 3.3 V supply rails
- Low noise: resistors kept in the hundreds of Ω (a 10 k Ω resistor adds ≈ 12.9 nV/VHz, far above the OPA1612's 1.1 nV/VHz)
- Small footprint
- C0G capacitors in the signal path

Topology Rationale

- **MFB over Sallen-Key:** MFB attenuates (gain = $R1/R3$ can be below 1) and subtracts in one op-amp; its stopband keeps rolling off into the MHz range where Sallen-Key feedthrough limits rejection to roughly 40–60 dB; its inputs sit at a virtual ground, so there is no common-mode input swing or common-mode distortion on ± 3.3 V rails.
- **Butterworth:** maximally flat passband with no ripple. A Bessel response at the same corner droops more at 20 kHz (≈ 0.3 dB vs ≈ 0.05 dB at a 60 kHz corner).

Input Levels (from I/V stage, $R_f = 499\ \Omega$)

Quantity	Nominal	Worst case
I/V output peak per leg	0.96 V	1.08 V
Differential input peak ($2 \times$ leg)	1.93 V	2.17 V

Component Values (TI labels)

Ref	Position	Value	Notes
R3	Input: I/V output $\rightarrow$ node X	499 Ω	Same value as I/V R_f (BOM consolidation)
R1	Feedback: output $\rightarrow$ node X	220 Ω	Gain $R1/R3 = 0.441$
R2	Middle: node X $\rightarrow$ (–) input	681 Ω	Chosen so both capacitors land within 1% of standard values
C1	Node X $\rightarrow$ ground	13 nF	One per X node (two per channel); calculated 12.89 nF. Also filters common-mode. C0G
C2	Feedback: output $\rightarrow$ (–) input	2.7 nF	Calculated 2.68 nF. C0G

The second half of each channel mirrors the first: its R1 goes to GND instead of the output, and its C2 goes from the (+) input to GND.

Components

Quantities cover both channels.

Ref	Part	Value	Tolerance	Package	Qty	Notes
U	TI OPA1612AIDRGR	Dual op-amp	N/A	8-SON 3 × 3 mm	1	One op-amp per channel
R3	YAGEO RT0402BRD07499RL	499 Ω, 1/16 W	±0.1%	0402	4	Thin film; same part as I/V R _f (library #1-105)
R1	YAGEO RT0402BRD07220RL	220 Ω, 1/16 W	±0.1%	0402	4	Thin film; feedback and mirrored ground leg (library #1-109)
R2	YAGEO RT0402BRD07681RL	681 Ω, 1/16 W	±0.1%	0402	4	Thin film (library #1-110)
C1	Murata GRM2195C1H133JA01D	13 nF, 50 V, C0G/NP0	±5%	0805	4	Library #1-108
C2	Murata GRM1555C1H272GE01D	2.7 nF, 50 V, C0G/NP0	±2%	0402	4	Feedback and mirrored ground leg (library #1-107)

Equations

Transfer function (TI labels):

$$H(s) = \frac{\frac{1}{R_2 R_3 C_1 C_2}}{s^2 + s\left(\frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) + \frac{1}{R_1 R_2 C_1 C_2}\right)}$$

Matching the denominator to $s^2 + (\omega_0/Q)s + \omega_0^2$:

$$C_1 = \frac{Q}{\omega_0} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \quad C_2 = \frac{1}{\omega_0^2 R_1 R_2 C_1} \quad \text{Gain} = \frac{R_1 R_3}{R_2}$$

Butterworth magnitude and passband loss:

$$|H|^2 = \frac{1}{1 + (f/f_0)^4} \quad \text{loss, (dB)} = 10 \log_{10} \left(1 + (f/f_0)^4 \right)$$

Results

Quantity	Value
f_o (standard values)	$\approx 69.4\text{ kHz}$
Q (standard values)	≈ 0.707
Summing-stage loss at 20 kHz	$\approx 0.030\text{ dB}$
Total chain loss at 20 kHz (with I/V $\approx 0.038\text{ dB}$)	$\approx 0.068\text{ dB}$
Summing-stage attenuation at 1 MHz (40 dB/decade)	$\approx 46\text{ dB}$
Total attenuation at 1 MHz (with I/V pole)	$\approx 60\text{ dB}$

Loss for $Q \neq 0.7071$ uses the general second-order form $|H|^{-2} = (1 - x^2)^2 + (x/Q)^2$, with $x = f/f_o$.

Simulation

Setup: SPICE AC sweep of one channel with the TI OPA1612-Q1 model on $\pm 3.3\text{ V}$. Differential drive: 0.5 V AC on each input, 180° apart (1 V differential). Logarithmic sweep, 10 Hz to 10 MHz , 100 points/decade. I/V stage not included.

Quantity	Simulated	Hand calculation
Passband gain	$\approx -7.1\text{ dB}$	-7.1 dB (0.441)
-3 dB frequency	69.30 kHz	$\approx 69.4\text{ kHz}$
Passband shape	Flat through 20 kHz , no peaking	$Q \approx 0.707$
Stopband slope	40 dB/decade	40 dB/decade
Level at 1 MHz	$\approx -53\text{ dB}$ ($\approx 46\text{ dB}$ below passband)	$\approx 46\text{ dB}$ below passband
High-frequency floor	Flattens near -84 dB around $5\text{--}10\text{ MHz}$	Not modeled

The flattening above about 5 MHz is the OPA1612 running out of gain-bandwidth (40 MHz); it remains about 77 dB below the passband.

Open Problems

- ☐ **Leg-to-input assignment.** The MFB inverts; choose which I/V leg feeds which input so overall polarity is correct.

Rev Log

Rev	Date	Change
3	2026-09-24	Components section added with part numbers. All resistors 0.1% thin film; resistor tolerance open problem closed.
2	2026-09-24	Simulation section added: AC sweep confirms -7.1 dB passband, -3 dB at 69.30 kHz, no peaking, 40 dB/decade roll-off.
1	2026-09-24	Standard values selected: $R2 = 681\ \Omega$, $C1 = 13\ \text{nF}$ to ground at each X node, $C2 = 2.7\ \text{nF}$ ($f_o \approx 69.4\ \text{kHz}$, $Q \approx 0.707$). Results updated. Standard capacitor values open problem closed.
0	2026-09-24	Initial release. Topology, requirements, input levels, component values, equations, results, open problems.

Sources

- [TI Analog Engineer's Circuit Cookbook: Amplifiers, MFB low-pass filter circuit](#)
- OPA1611/OPA1612 datasheet, Audio DAC Post Filter (I/V Converter and Low-Pass Filter) application circuit
- [OPA1612 product page, TI](#)